Exercise for chapter 10

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Chapter 10

10.1 CaF_2 and MgF_2 are mutually insoluble in the solid state and form a simple binary eutectic system. Calculate the composition and temperature of the eutectic melt assuming that the liquid solutions are Raoultian. The actual eutectic occurs at $X_{CaF_2} = 0.45$ and T = 1243 K.

Solution to 10.1

$$\Delta G^{\circ}_{m(A)} = -RT \ln a_A$$

if the liquid solutions are Raoultian,

$$\Delta G^{\circ}_{m(A)} = -RT \ln X_A$$

Let $x = X_{CaF_2}$

For the CaF_2 liquid line

$$\Delta G^{\circ}_{M} = 31200 - 18.45T = -8.3144T \ln x$$

For the $\mathrm{Mg}F_2$ liquid line

$$\Delta G^{\circ}_{M} = 58160 - 37.86T = -8.3144T \ln(1-x)$$

Simultaneous solution gives the point of intersection of the liquid lines at

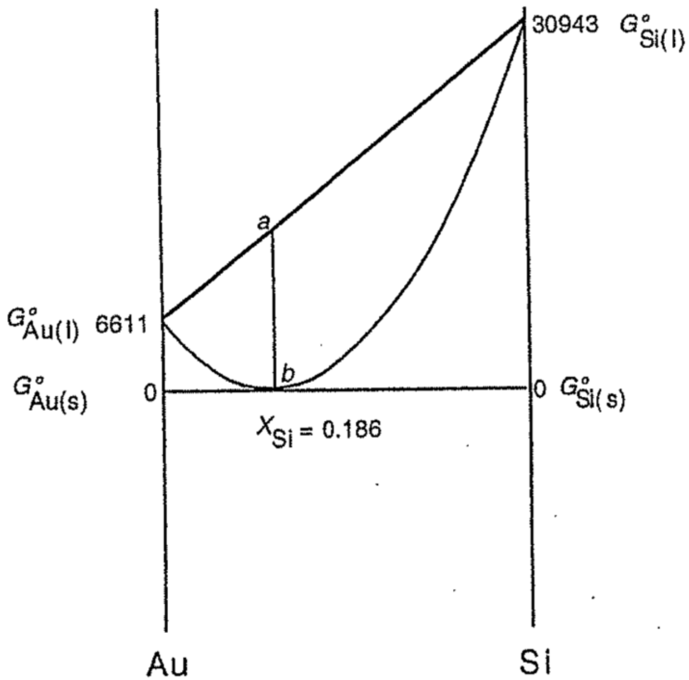
$$T = 1317K \qquad X_{CaF_2} = 0.53$$

The actual eutectic is

$$T = 1243K \qquad X_{CaF_2} = 0.45$$

10.2 Gold and silicon are mutually insoluble in the solid state and form a eutectic system with a eutectic temperature of 636 K and a eutectic composition of $X_{Si} = 0.186$. Calculate the Gibbs free energy of the eutectic melt relative to (a)unmixed liquid Au and liquid Si, and (b) unmixed solid Au and solid Si.

Solution to 10.2



$$\Delta G^\circ_{m(A)} = \Delta H^\circ_{m(A)}\left(\frac{T_{m(A)} - T}{T_{m(A)}}\right)$$

At 636K

$$\Delta G^\circ_{m,Au} = 12600\left[\frac{1338 - 636}{1338}\right] = 6611\,J$$

and

$$\Delta G^\circ_{m,\mathrm{Si}} = 50200\left[\frac{1685 - 636}{1685}\right] = 30943\,J$$

a).

$$\Delta G^M = ab = -(0.186 \times 30943 + 0.814 \times 6611)J$$

$$= -11140\,J$$

b). $\Delta G^M = 0$

Chapter 10

10.3 Al_2O_3, which melts at 2324 K, and Cr_2O_3, which melts at 2538 K, form complete ranges of solid and liquid solutions. Assuming that $\Delta S^{\circ}_{m,Al_2O_3} = \Delta S^{\circ}_{m,Cr_2O_3}$ and that the solid and liquid solutions in the system Al_2O_3 - Cr_2O_3 behave ideally, calculate

a. The temperature at which equilibrium melting begins when an alloy of $X_{Al_2O_3} = 0.5$. is heated

b. The composition of the melt which first forms

c. The temperature at which equilibrium melting is complete

d. The composition of the last-formed solid

Solution to 10.3

$$\frac{\Delta G^{\circ}_{m,Al_2O_3}}{RT} = \frac{12929}{T} - 5.563$$

$$\frac{\Delta G^{\circ}_{m,Cr_2O_3}}{RT} = \frac{14119}{T} - 5.563$$

$$X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}{\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right) - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}$$

a) The solidus temperature T, at $X_{Al_2O_3} = 0.5$ is given by

$$\frac{1 - exp\left(-\frac{14119}{T} + 5.563\right)}{exp\left(-\frac{12929}{T} + 5.563\right) - exp\left(-\frac{14119}{T} + 5.563\right)} = 0.5$$

Which has the solution $T = 2418\ K$

Solution to 10.3

b) The corresponding liquidus composition is

$$X_{Al_2O_3}(liquidus) = 0.5\left[exp\left(-\frac{12929}{2418}+5.563\right)\right] = 0.621$$

c) The liquidus temperature T, at $X_{Al_2O_3} = 0.5$ is given by

$$\frac{1-exp\left(-\frac{14119}{T}+5.563\right)}{exp\left(-\frac{12929}{T}+5.563\right)-exp\left(-\frac{14119}{T}+5.563\right)}$$

$$\times\ exp\left(-\frac{12929}{T}+5.563\right) = 0.5$$

$$X_{A(l)} = \frac{\left[1-\exp\left(-\frac{\Delta G^\circ_{m(B)}}{RT}\right)\right]\exp\left(-\frac{\Delta G^\circ_{m(A)}}{RT}\right)}{\exp\left(-\frac{\Delta G^\circ_{m(A)}}{RT}\right)-\exp\left(-\frac{\Delta G^\circ_{m(B)}}{RT}\right)}$$

Which has the solution $T = 2444\ K$

Solution to 10.3

d) The corresponding solidus composition is

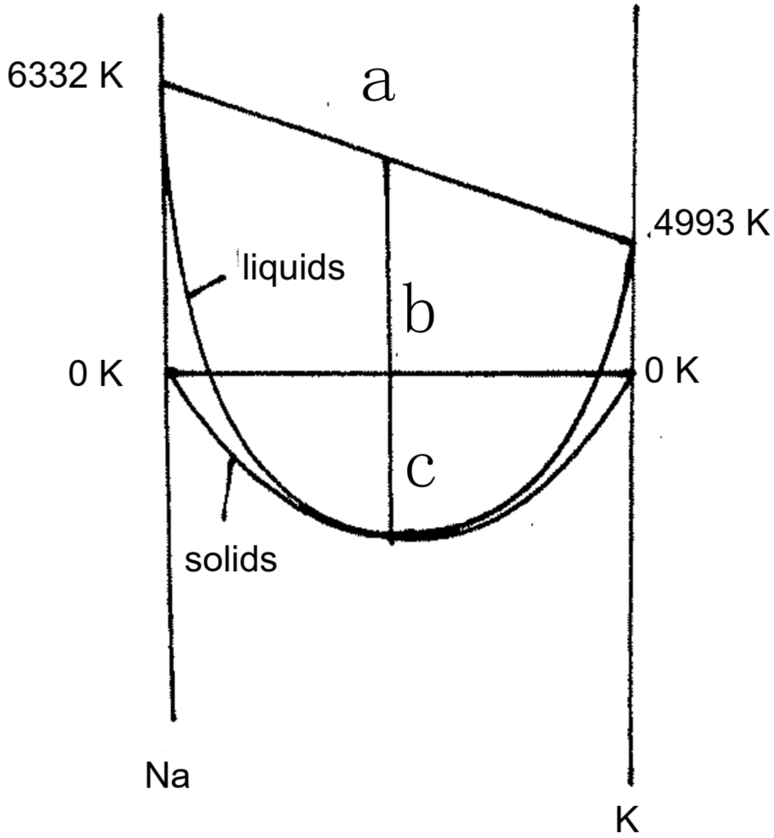
$$X_{A(s)} = \frac{1-\exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}{\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right)-\exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}$$

$$X_{Al_2O_3}(\text{solidus}) = \frac{1-exp\left(-\frac{14119}{2444}+5.563\right)}{exp\left(-\frac{12929}{2444}+5.563\right)-exp\left(-\frac{14119}{2444}+5.563\right)}$$

$$= 0.38$$

10.4 $Na_2O.B_2O_3$ and $K_2O.B_2O_3$ form complete ranges of solid and liquid solutions and the solidus and liquidus show a common minimum at the equimolar composition and $T = 1123\ K$. Calculate the molar Gibbs free energy of formation of the equimolar solid solution from solid $Na_2O.B_2O_3$ at 1123 K, assuming that the liquid solutions are ideal.

Solution to 10.4



Solution to 10.4

$$\Delta G^{\circ}_{m(A)} = \Delta H^{\circ}_{m(A)} \left(\frac{T_{m(A)} - T}{T_{m(A)}} \right)$$

At $T = 1123\ K$

$$\Delta G^{\circ}_{m,Na_2O.B_2O_3} = 67000 \left(\frac{1240 - 1123}{1240} \right) = 6322\ J$$

$$\Delta G^{\circ}_{m,K_2O.B_2O_3} = 62800 \left(\frac{1220 - 1123}{1220} \right) = 4993\ J$$

$$\Delta G^{M}(flow\ liquids) = 8.3144 \times 1123[0.5 \ln 0.5 + 0.5 \ln 0.5] = -6472J$$

$= ac$

$$\Delta G_s^M = RT(X_A \ln X_A + X_B \ln X_B)$$

$$ab = 0.5 \times 6322 + 0.5 \times 4993 = 5658\ K$$

So $\Delta G^M = bc = -6472 + 5658 = -814J$

10.5 SiO_2, which melts at 1723°C, and TiO_2, which melts at 1842°C, are immiscible in the solid state, and the $SiO_2 - TiO_2$ binary system contains a monotectic equilibrium at 1794°C, at which temperature virtually pure TiO_2 is in equilibrium with two liquids containing mole fractions of SiO_2 of 0.04 and 0.76. If, for the purpose of simple calculation, it is assumed that the compositions of the two liquids are $X_{SiO_2} = 0.24$ and $X_{SiO_2} = 0.76$ and that the liquid solutions are regular in their behavior, what is the value of α_l and at what temperature does the liquid immiscibility gap disappear?

Solution to 10.5

At the composition of the double tangent to the ΔG^M curve

$$\Delta G^M = RT(X_A \ln X_A + X_B \ln X_B) + \alpha X_A X_B$$

$$\frac{d\Delta G^M}{dx} = RT \ln\left(\frac{x}{1-x}\right) + \alpha_l(1-2x) = 0$$

At $T = 2067\ K$

$$8.3144 \times 2067 \ln\left(\frac{0.24}{1-0.24}\right) + \alpha_l(1 - 2 \times 0.24) = 0$$

Which gives $\alpha_l = 38096\ J$

$$T_{cr} = \frac{\alpha}{2R} \qquad T_{Cr} = \frac{\alpha_l}{2R} = \frac{38096}{2 \times 8.3144} = 2291\ K$$

10.6 The binary system Ge–Si contains complete solid and liquid solutions. The melting temperatures are $T_{m,Si} = 1685\ K$, and $T_{m,Ge} = 1210\ K$, and $\Delta H^{\circ}_{m,Si} = 50200\ J$.At 1200°C, the liquidus and solidus compositions are, respectively, $X_{Si} = 0.32$ and $X_{Si} = 0.665$. Calculate the value of $\Delta H^{\circ}_{m,Ge}$, assuming that

a. The liquid solutions are ideal.

b. The solid solutions are ideal.

Which assumption gives the better estimate? The actual value of $\Delta H^{\circ}_{m,Ge}$ at $T_{m,Ge}$ is 36,900 J.

Solution to 10.6

At $T = 1473\ K$

$$\Delta G^{\circ}_{m(A)} = \Delta H^{\circ}_{m(A)}\left(\frac{T_{m(A)} - T}{T_{m(A)}}\right) \qquad X_{A(s)} = \frac{1 - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}{\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right) - \exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}$$

$$\Delta G^{\circ}_{m,Si} = 50200\left(\frac{1685 - 1473}{1685}\right) = 6316\ J$$

$$\frac{\Delta G^{\circ}_{m,Si}}{RT} = \frac{6316}{8.3144 \times 1473} = 0.5157$$

$$X_{Ge}(\text{solidus}) = \frac{1 - exp(-0.5157)}{exp\left(-\frac{\Delta G^{\circ}_{m,Ge}}{8.3144 \times 1473}\right) - exp(-0.5157)} = 0.335$$

Which gives $\Delta G^{\circ}_{m,Ge} = -7197\ J$ $\quad \Delta H^{\circ}_{m,Ge} = \dfrac{-7197}{\left(\frac{1210 - 1473}{1210}\right)} = 33111\ J$

$$X_{A(l)} = \frac{\left[1-\exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)\right]\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right)}{\exp\left(-\frac{\Delta G^{\circ}_{m(A)}}{RT}\right)-\exp\left(-\frac{\Delta G^{\circ}_{m(B)}}{RT}\right)}$$

Solution to 10.6

$$X_{Ge}(\text{liquidus}) = \frac{[1-exp(-0.5157)]exp\left(-\frac{\Delta G^{\circ}_{m,Ge}}{8.3144 \times 1473}\right)}{exp\left(-\frac{\Delta G^{\circ}_{m,Ge}}{8.3144 \times 1473}\right) - exp(-0.5157)} = 0.68$$

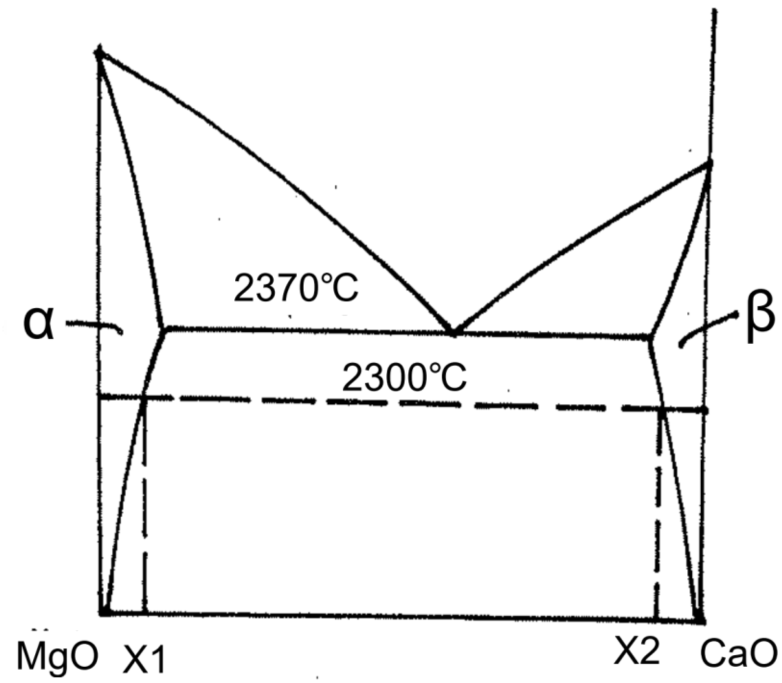
Which gives $\Delta G^{\circ}_{m,Ge} = -4679\ J$

$$\Delta H^{\circ}_{m,Ge} = \frac{-4679}{\left(\frac{1210-1473}{1210}\right)} = 21527\ J$$

So The estimate from the solidus(33100 J) is closer to the actant value of 36900 J

10.7 CaO and MgO form a simple eutectic system with limited ranges of solid solubility. The eutectic temperature is 2370°C. Assuming that the solutes in the two solid solutions obey Henry's law with $\gamma^{\circ}_{\mathrm{CaO}}$ in MgO = 12.88 and $\gamma^{\circ}_{\mathrm{MgO}}$ in CaO= 6.23 at 2300°C, calculate the solubility of CaO in MgO and the solubility of MgO in CaO at 2300°C.

Solution to 10.7



$$a_{CaO}(in\ \beta\ at\ X_2) = a_{CaO}(in\ \alpha\ at\ X_1)$$

$$1 - X_1 = 6.23(1 - X_2) \qquad \text{(i)}$$

$$a_{\mathrm{Mg}O}(in\ \alpha\ at\ X_1) = a_{\mathrm{Mg}O}(in\ \beta\ at\ X_2)$$

$$12.88X_1 = X_2 \qquad \text{(ii)}$$

Solution of (i) and (ii) gives

$$X_1 = X_{CaO} = 0.066$$

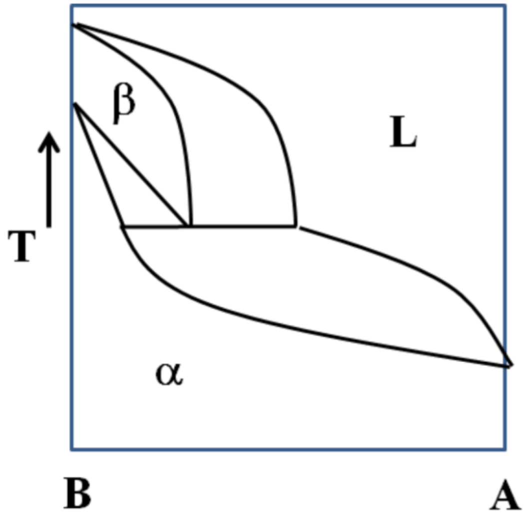
$$X_2 = X_{\mathrm{Mg}O} = 0.85$$

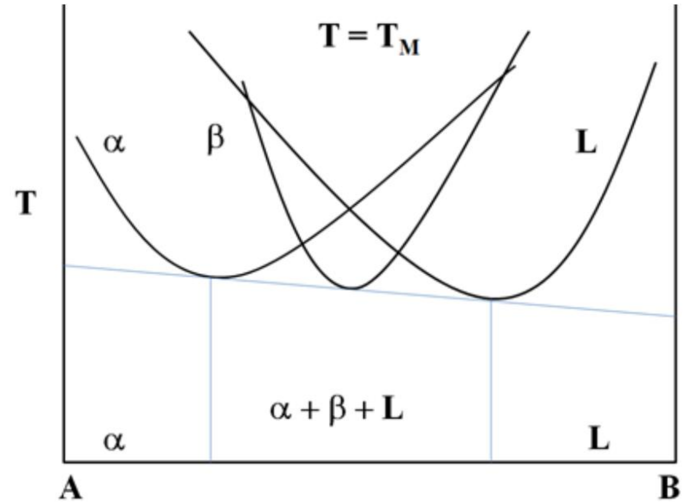
10.8 A metatectic binary phase diagram displays the following invariant transformation on cooling:

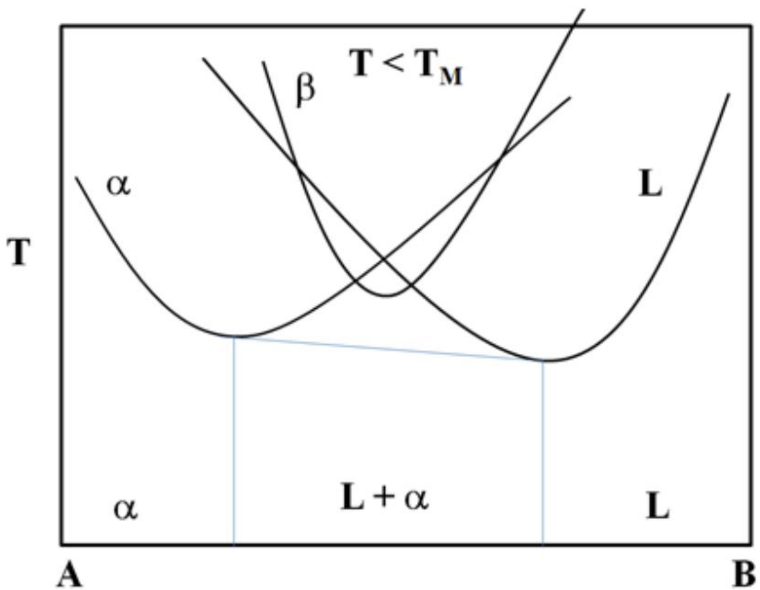
$$\beta \longrightarrow \alpha + L$$

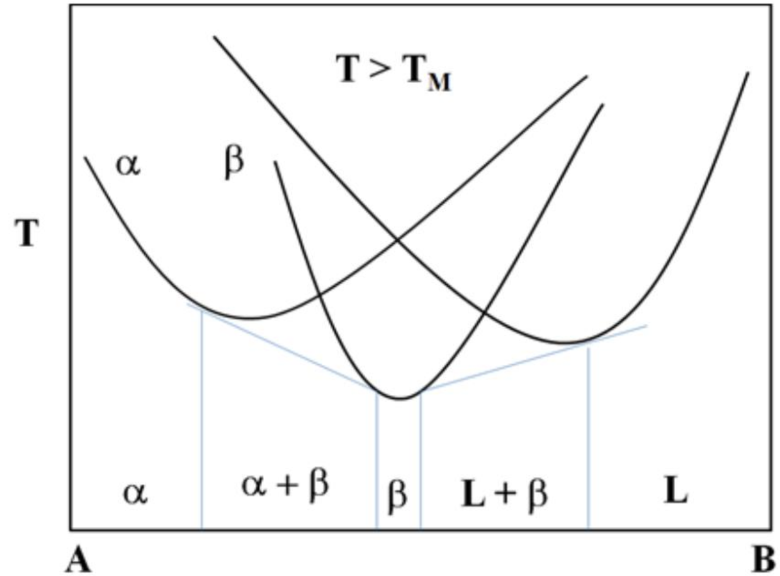
Sketch such a phase diagram and then draw free energy curves of mixing just below, at, and just above the invariant temperature.

Solution to 10.8









10.9 The free energy of mixing of a regular solution is given by

$$\Delta G_{mix} = \alpha X_A X_B + RT(X_A \ln X_A + X_B \ln X_B)$$

$$\alpha = 24943\ J/mole$$

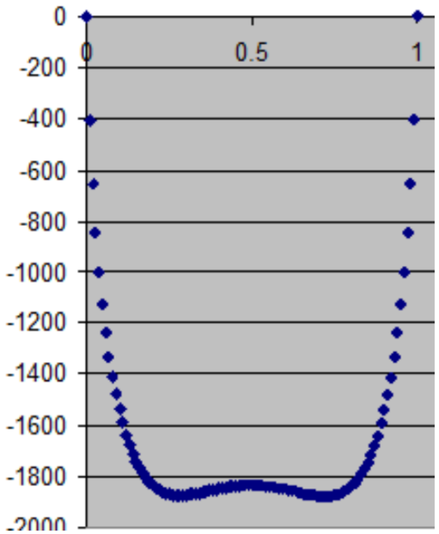
a. Plot ΔG_{mix} versus X_B at 1400, 1500, and 1600 K.

b. Plot $\partial\Delta G_{mix}\ /\ \partial X_B$ versus X_B at the same temperatures as (a).

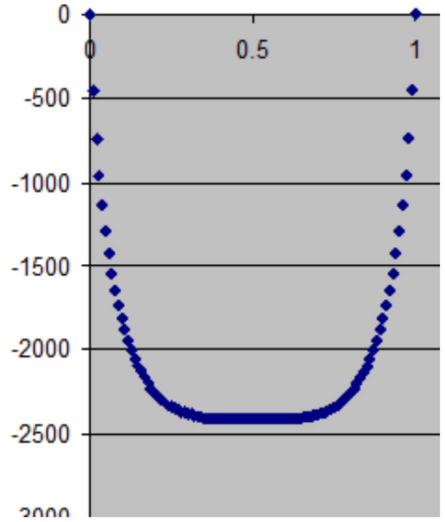
c. Determine the critical temperature for this alloy. Show your work.

Solution to 10.9

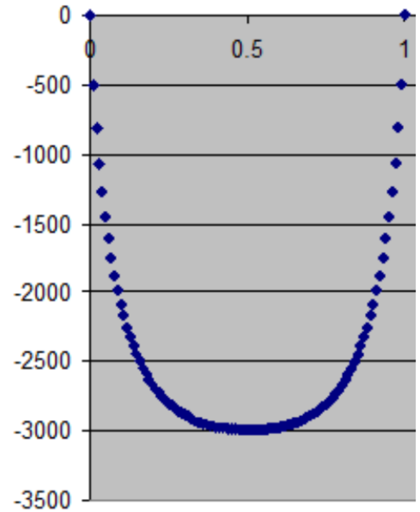
a)



1400 K



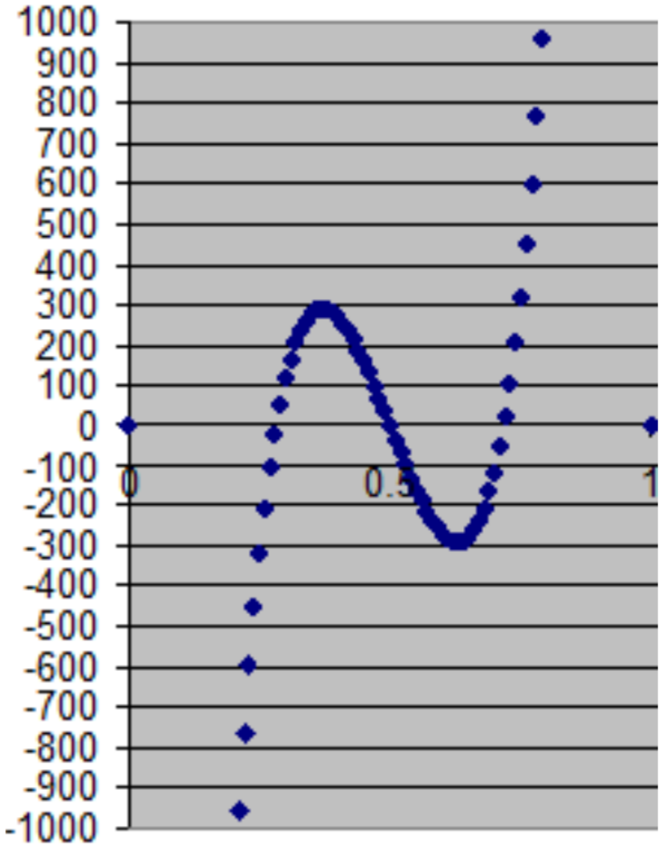
1500 K



1600 K

Solution to 10.9

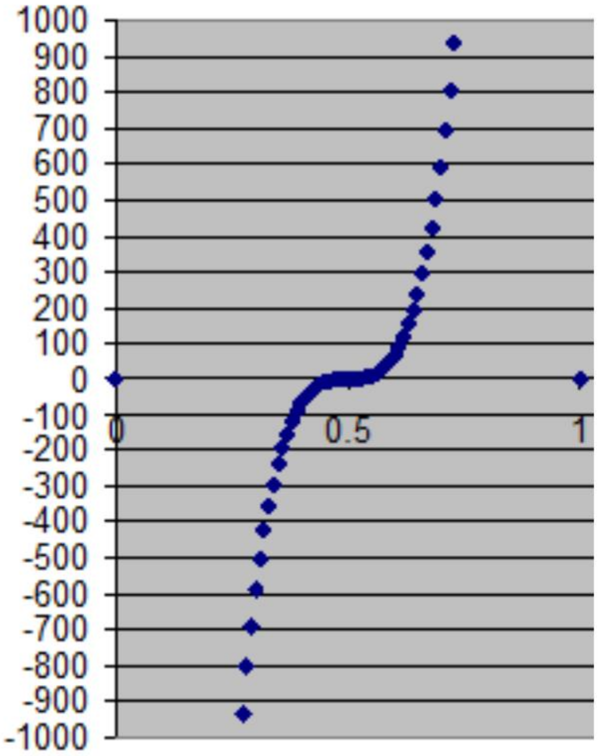
b)



Plot of the first derivative for 1400K. The first and third zeros are the minima of free energy plot: they give the compositions of the two phases in equilibrium. The maximum in this plot and the minimum give us the spinodals.

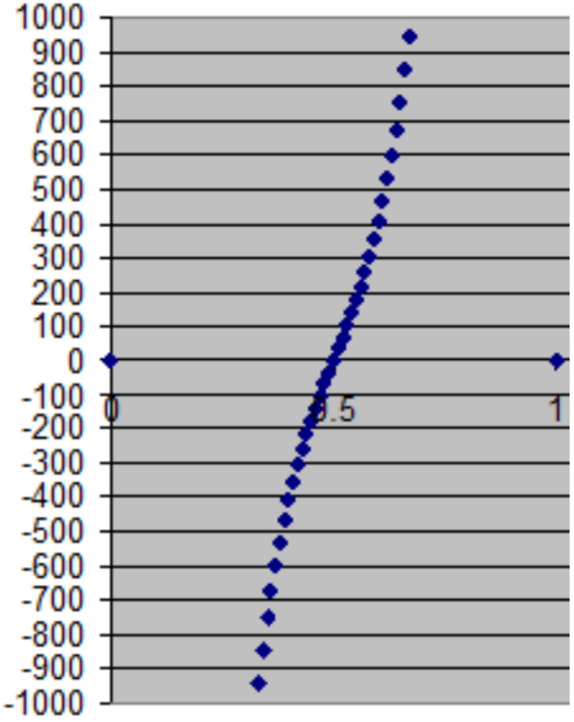
Solution to 10.9

b)



Plot of the first derivative for 1500K. Only one zero for this plot at 0.5. In fact when it passes through zero the curve has zero slope. This demonstrates that this is the critical temperature

Solution to 10.9

b) 

Plot of the first derivative for 1600K.The zero value is the minimum in free energy vs. X_B plot (0.5)

c) From the answers above we estimate T_C to 1500K. Or, since a regular solution we calculate it to be $T_C = \frac{a_0}{2R} = 1500K$

10.10 A certain solid solution of A and B has a ΔG_{mix}^{XS} as follows:

$$\Delta G_{mix}^{XS} = \Delta G_{mix} - \Delta G_{mix}^{id} = X_A X_B (a_1 X_A + a_2 X_B) + RT(X_A \ln X_A + X_B \ln X_B)$$

Where:

a_1 = 12,500 J/mole

a_2 = 5,500 J/mole

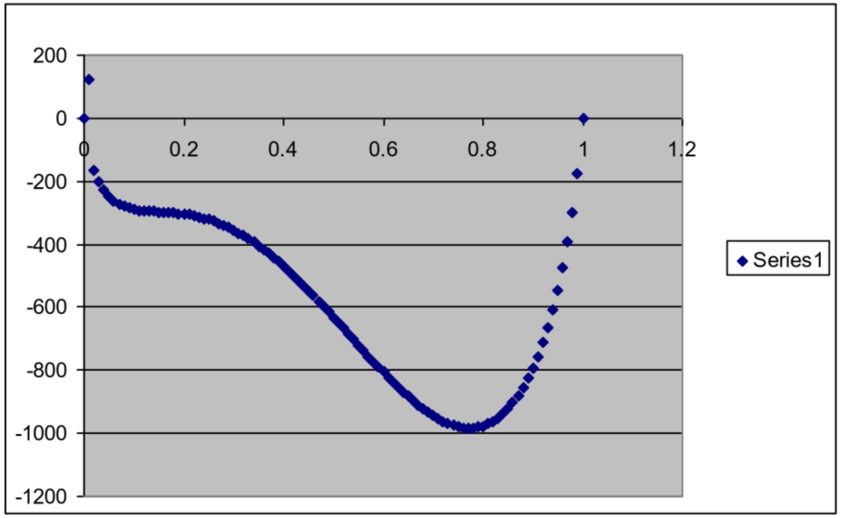
a. Plot ΔG_{mix}^{XS} at T = 500 and 700 K.

b. Sketch the T versus X_B phase diagram of this alloy.

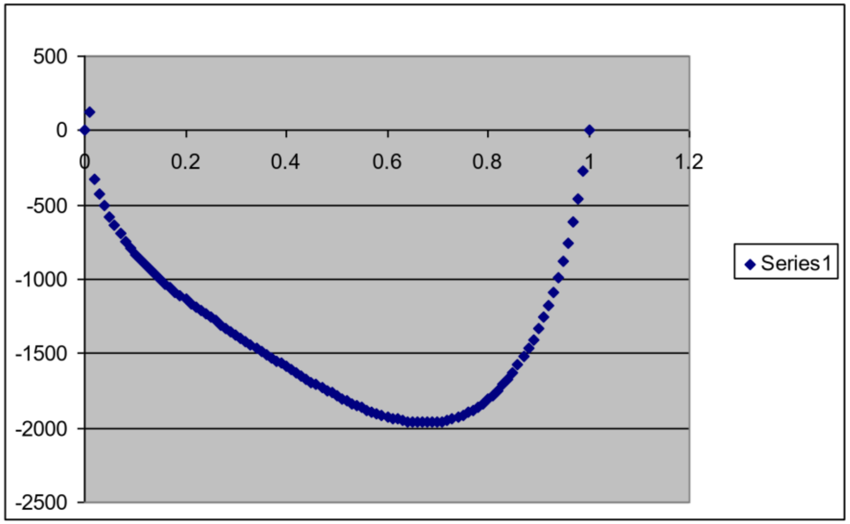
c. Determine the critical temperature and composition of this alloy.

Solution to 10.10

a)



Plot of ΔG_{mix}^{XS} vs. X_B at 500 K, showing that a miscibility gap will exist in the T vs. X_B phase diagram with TC being less than $X_B = 0.5$.



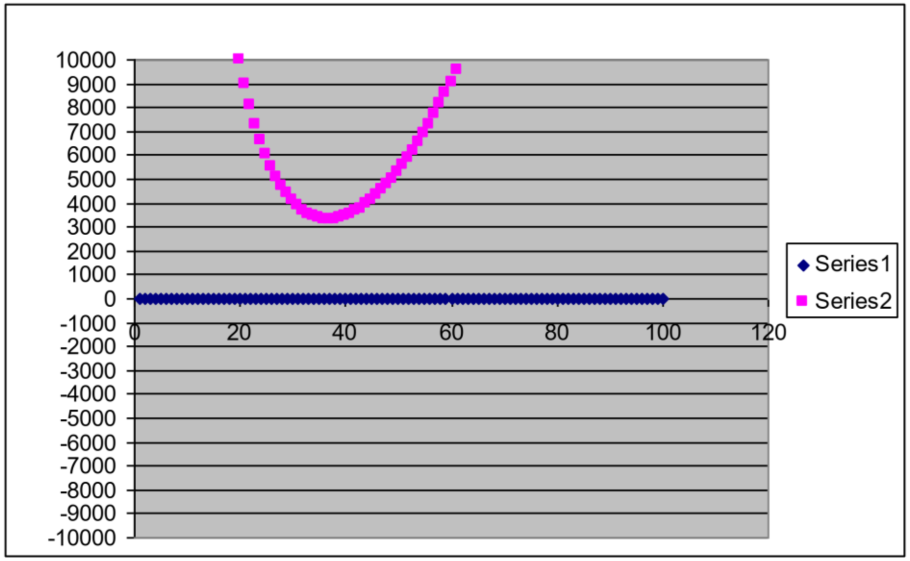
Plot of ΔG_{mix}^{XS} vs. X_B at 700K, showing that 700 K is above T_C .

Solution to 10.10

b) Plot will be asymmetric with T_C less than $X_B = 0.5$.

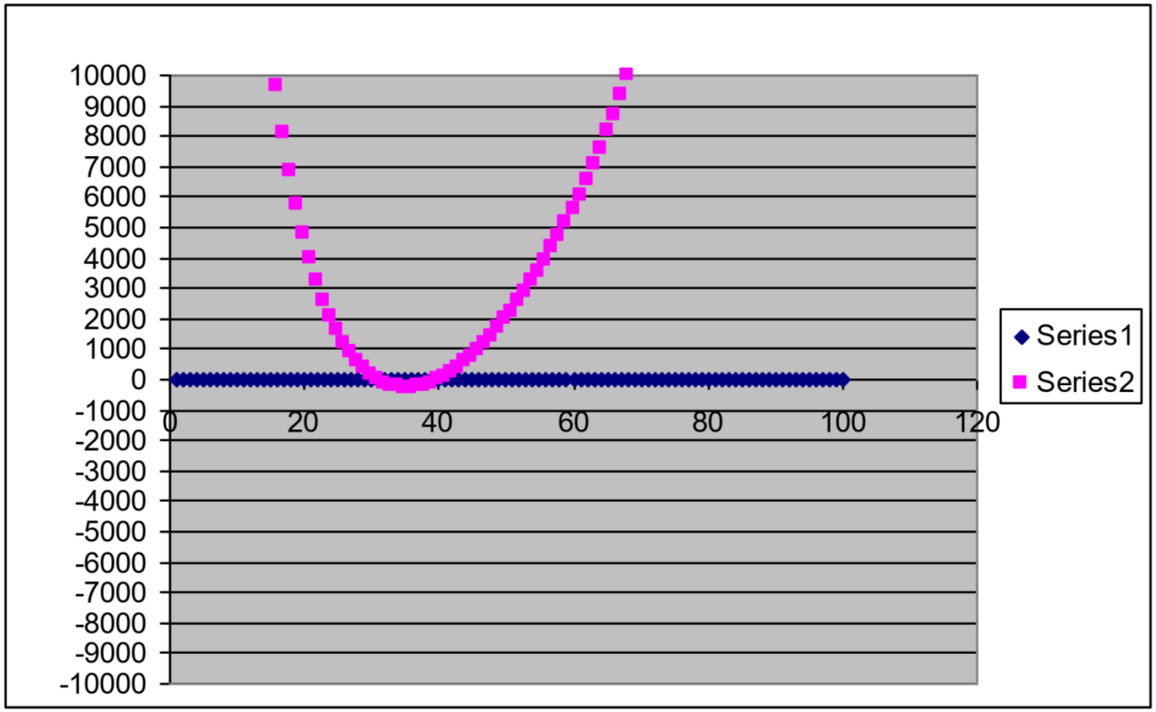
Solution to 10.10

c) A graphical way of doing this is to plot $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial_{X_B}^2}$ vs. X_B and see what T has the curve just touching the 0 ordinate. Below is the $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial_{X_B}^2}$ vs. X_B for 700 K .



This is above T_C.The 600K plot shows two crossings of the $\frac{\partial^2 \Delta G_{mix}^{XS}}{\partial_{X_B}^2} = 0$. Hence $T < T_C$.

Solution to 10.10



By trying various values of T, T_C can be found to be 607.5 K and the composition of the critical point is $X_B = 0.35$.

10.11 The phase diagram of an alloy that has a eutectoid transformation is shown in Figure 10.33.

a. Sketch the Gibbs free energy curves for this alloy at T = T'.

b. Show that the α/γ solvus must enter the α/β two-phase field (as shown by the arrow)

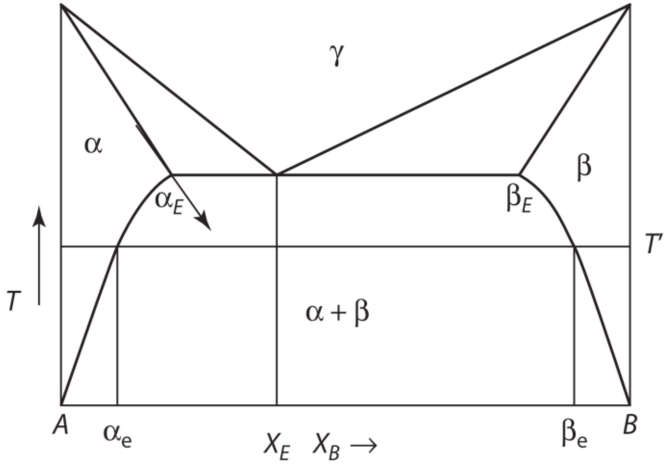
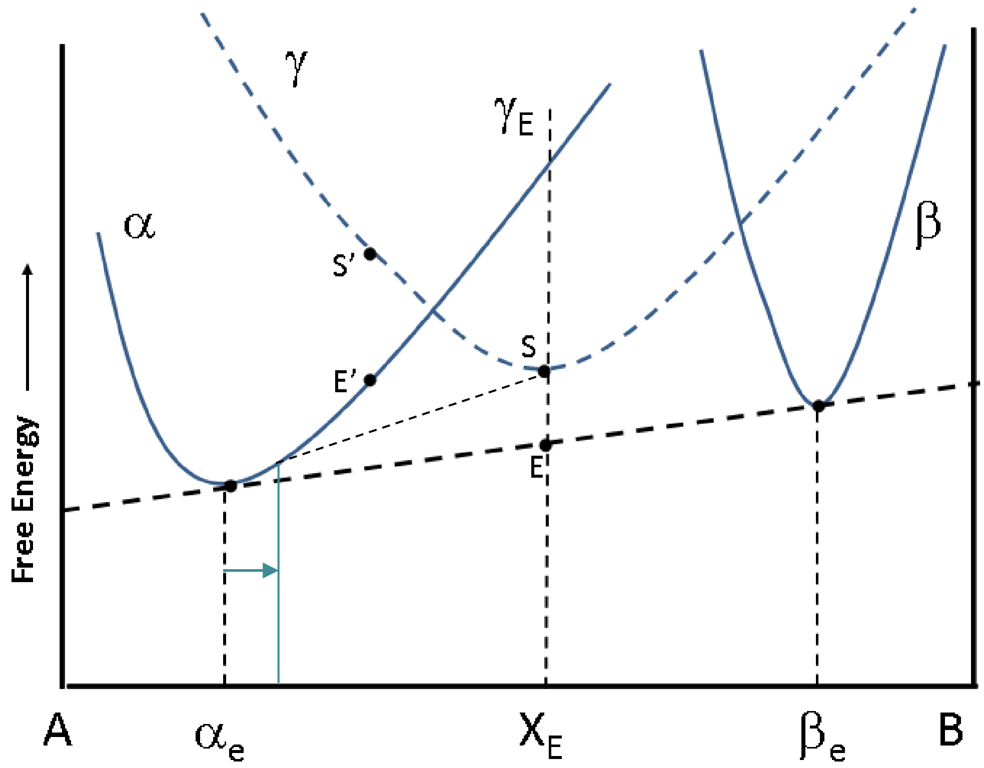


Figure 10.33: A eutectoid phase diagram showing the extension of the α/γ equilibrium curve into the α/β solvus.

Solution to 10.11

a)

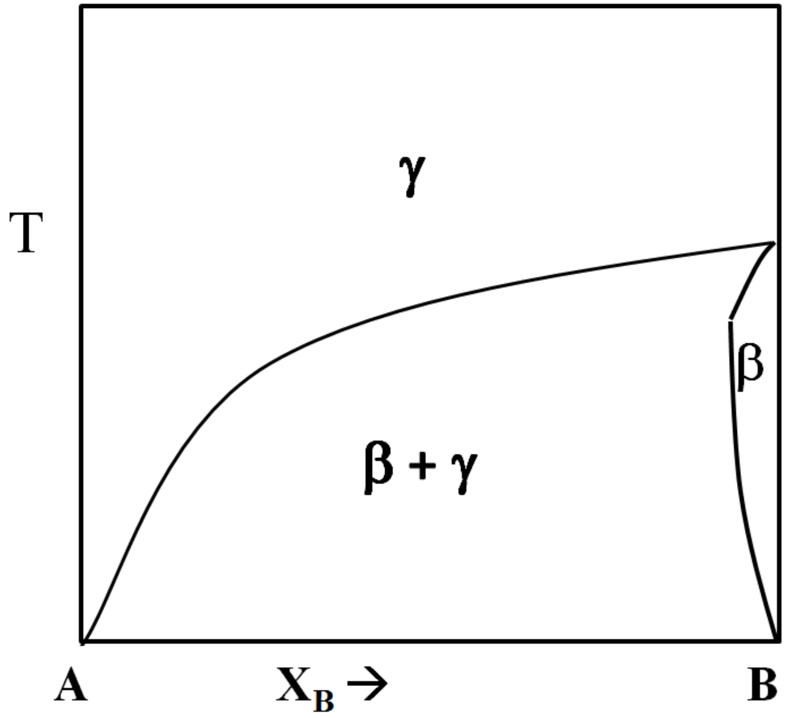


b) The common tangent of γ and α below T_E intersects the α free energy curve to the right of α_e.

10.12 A eutectoid phase diagram is shown in Figure 10.33. Draw the phase diagram that would result if for some reason it were impossible to form the α phase.

Solution to 10.12

A possible diagram is shown below. Note that the γ/β curve is the same as the one in Fig.10.28a until the temperature where the eutectoid temperature was. is reached. It then is extrapolated to 0 K. The 0 K phases present are γ and β .



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